

Manual for multi-target tracking with towed hydrophone arrays

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To all the whale nerds out there ;)

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Chapter 1

Overview

1.1 About this manual

This manual guides you through a workflow for tracking multiple underwater acoustic sources along a straight trackline with towed hydrophone arrays. The workflow consists of three packages (one of them optional), and each is discussed in detail in the subsequent chapters.

In this chapter the setup and required packages are outlined in Section 1.2, the overview of the workflow is given in Section 1.3, and the expected folder structure is outlined in Section 1.4.

1.2 Setup, required toolboxes and packages

The packages are developed in Matlab, and tested on **Matlab version 2022a (9.12)**.

IMPORTANT: Do not use earlier versions of Matlab since certain functions have changed behavior. For example, function `ismembertol` (used in package “Localization towed array”- Chapter 4) has different behavior between version 2022a and older versions, and the processing will not work correctly if using older versions. Also, earlier versions of Matlab do not have certain functions such as `clim` and `FontSize` (used for plotting in package “Localization towed array”- Chapter 4), so these will not work. **Also, do not use Matlab version 2023b (23.2.0.2485118)** since function `ismembertol` also changed behavior and thus “Localization towed array” package will not work correctly.

Three packages are available:

- 1) “**TDOA tracking master**” package (Chapter 2): A Matlab package for multi-target tracking of Time-Differences-Of-Arrivals (TDOAs) of signals between two sensors from towed hydrophone array recordings.

GitHub link: https://github.com/PinaGruden/TDOA_tracking_master.

- 2) “**Extract Pamguard detections**” package (Chapter 3) [Optional]: A Matlab package for extracting detection information from SQLite3 database produced

by open source software Pamguard into Matlab tables.

GitHub link: https://github.com/PinaGruden/Extract_Pamguard_detections.

- 3) “**Localization towed array**” package (Chapter 4): A Matlab package for localizing an acoustic group and computing perpendicular distance to boat track-line.

GitHub link: https://github.com/PinaGruden/Localization_Towed_Array.

Required Matlab toolboxes for each package:

- 1) “**TDOA tracking master**” package:
 - Signal Processing Toolbox
 - Statistics and Machine Learning Toolbox
- 2) “**Extract Pamguard detections**” package:
 - Statistics and Machine Learning Toolbox
- 3) “**Localization towed array**” package:
 - Signal Processing Toolbox
 - Mapping Toolbox
 - Image Processing Toolbox

Data Requirements for each package:

- 1) “**TDOA tracking master**” package:
 - The expected data format are `.wav` files that contain at least two channels. The expected name for `.wav` files is “xxx_yyyyMMdd_HHmmss_SSS.wav”, where “xxx_” can be any string (or none), and “yyyymmdd_HHmmss_SSS” part of the name specifies the date (year, month, day) and time (hours, minutes, seconds, and milliseconds (SSS)).
- 2) “**Extract Pamguard detections**” package:
 - The expected data format are `.sqlite3` database files that are produced by Pamguard.
- 3) “**Localization towed array**” package:
 - The expected data format are `.mat` files produced by “TDOA tracking master” and “Extract Pamguard detections” packages;
 - Optionally also `.csv` file containing GPS data

NOTE, the localization algorithm assumes that the boat moves in a straight line and no turns are present. If you have data with turns only process the part where boat travels in a straight line.

1.3 Workflow

Localization of multiple sources workflow is outlined in Figure 1.1.

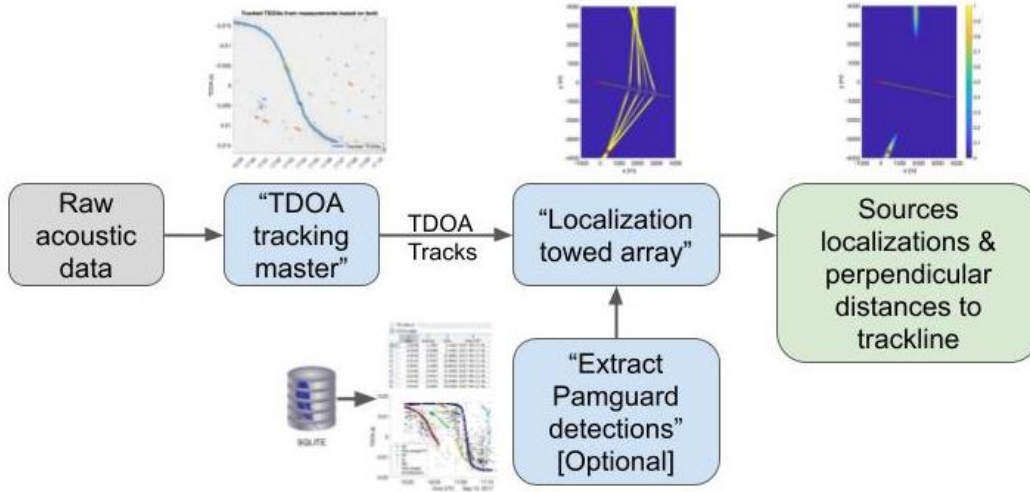


Figure 1.1: Diagram of the localization workflow with the packages outlined in this manual. The three packages discussed in this manual are color-coded in blue, and the output is color-coded in green. “TDOA tracking master” package performs tracking in the TDOA domain. “Localization towed array” package enables tracking in the spatial domain. “Extract Pamguard detections” package is optional, and it enables raw data to be imported from Pamguard’s .sqlite3 database.

1.4 Expected folder structure

The way the packages are organized is to run them in sequence (i.e., tracking in TDOA domain - “TDOA tracking master” package, should be done before one attempts to localize in the spatial domain - “Localization towed array” package) - see Figure 1.1.

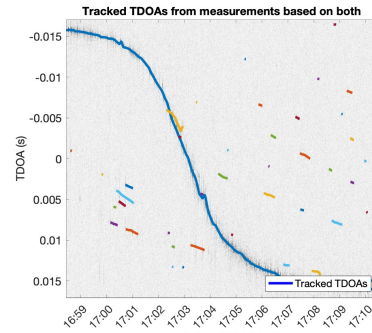
It is expected that you will be processing data from one encounter at a time, keeping data from each encounter separate. It is also expected that each encounter will be in a separate folder and have its results (and intermediate files) saved in a separate folder from other encounters. So for **each encounter** one should have:

- a separate folder for raw data (.wav files) - used in “TDOA tracking master”
- a separate folder for raw cross-correlogram - used in “TDOA tracking master” and “Localization towed array”
- a separate folder for results (Extracted TDOA tracks) - used in “TDOA tracking master” and “Localization towed array”
- (Optional) a separate folder for SQLite database - used in “Extract Pamguard detections”

- (Optional) a separate folder for extracted csv tables - used in “Extract Pamguard detections”
- (Optional) a separate folder for results (Pamguard detections) - used in “Extract Pamguard detections” and “Localization towed array”
- a separate folder for raw GPS data - used in “Localization towed array”
- a separate folder for extracted table from GPS data - used in “Localization towed array”
- a separate folder for Final localization results - used in “Localization towed array”

Note, most of these folders (apart from essential folders holding your data) will be automatically created on a path you specify in individual package if they do not already exist.

Chapter 2



“TDOA tracking master” package

2.1 Package overview

This is a Matlab based package that performs tracking of Time-Difference-Of-Arrivals (TDOAs) from multiple sources based on information from two moving sensors in a linear array. It was written and tested for towed arrays, but could also be used with stationary arrays.

The TDOAs are obtained from cross-correlograms, which are computed based on either clicks, whistles, or both. In addition to TDOAs, amplitudes are also obtained from the cross-correlograms. Both TDOAs and amplitudes represent measurements based on which the tracking is carried out. The tracking uses the GM-PHD-SA filter [1] to track (i.e., connect all TDOAs from one source or closely spaced group of sources into a continuous trajectory over time). See [1] for more details on the cross-correlogram computation, extraction of measurements, and tracking.

This package was developed in Matlab, and tested on **Matlab version 2022a (9.12)**.

2.2 Package contents

Package contains folders and functions that execute pre-processing, measurement extraction, and TDOA tracking.

The package contains the following functions, scripts, and files in the main folder:

- **A1_Compute_CrossCorrelograms.m** - main script that computes cross-correlograms.
- **A2_ExtractMeasurements_and_TrackTDOAs.m** - main script that extracts measurements from cross-correlograms and extracts TDOA tracks from them.
- **Array_Info.csv** - a table specifying array spacing.
- **plot_results.m** - plots TDOA tracking package results.
- **README.md** - readme document specifying package usage.
- **Specify_Parameters4Tracking.m** - specifies parameters for measurement extraction and tracking.

- **Specify_Parameters4Xcorr.m** - specifies parameters for cross-correlogram computation.
- **Specify_Paths.m** - specifies paths to folders where data is located and results are saved.

The package contains the following folders in the main folder:

- 1) **./Preprocess_Extract_Measurements/** - contains code to pre-process audio files, to construct cross-correlograms, and to extract measurements for tracking. Functions included are:
 - **compute_crosscorrelogram.m** - computes a cross-correlogram between two channels in audio data.
 - **extract_measure_crosscorrelogram.m** - extracts TDOAs and amplitudes of the cross-correlation information from a cross-correlogram and saves it as a random finite set (RFS).
 - **extract_peaks_crosscorr.m** - extracts peaks from the cross-correlogram.
 - **gcc.m** - computes the generalized cross-correlation (GCC) of two signals.
 - **getStrDateTime.m** - gets date and time from a file header.
 - **norm_background_crosscorr.m** - normalizes background noise to have std=1 under the Rayleigh assumption.
 - **preprocess.m** - applies de-clicking of the signal.
- 2) **./GMPHD_SA/** - contains code to extract TDOA tracks from cross-correlograms using the GM-PHD-SA filter. Functions and files included are:
 - **birth_process_amplitude.m** - generates newborn targets based on the measurements.
 - **gen_tdoa_tracking_models.m** - generates models required for TDOA tracking with the GM-PHD-SA filter.
 - **gmphd_adaptive_amplitude.m** - tracks multiple targets using the GM-PHD-SA filter.
 - **gmphd_limit.m** - eliminates Gaussian components based on their weights.
 - **gmphd_merge.m** - merges Gaussian components that are close together.
 - **gmphd_prune.m** - performs pruning of Gaussian components.
 - **kalman_predict_multiple.m** - uses a Kalman filter to predict target states according to system model.
 - **kalman_update_multiple.m** - uses a Kalman filter to update target states according to measurement model.
 - **kalman_update_single.m** - uses a Kalman filter to update a single target state according to measurement model.
 - **postprocess.m** - removes short tracks and smooths the tracks.
 - **tracktarget_tdoa_labels.m** - finds targets with the same label and collects them into tracks.

- BayesOptimization_GMPHDPParams_GTchunked_lambda4.5_DiffBirth.mat - optimized parameters.
 - BirthVelocityPrior_AllTrainData.mat - birth velocity prior from training data.
- 3) ./Test_example/ - contains data to use for testing the package works correctly.

2.3 How to use

As outlined in 1.4, the package expects a certain folder structure, that is for convenience outlined in 2.3.1. In order to use this package you must first specify the array used to record the measurements and parameters that you want to use for tracking - see Section 2.3.2. After this, you can run the measurements extraction and tracking - see Section 2.3.3. The package outputs are outlined in Section 2.4.

2.3.1 Expected folder structure

It is expected that each encounter will be in a separate folder and have its results (and intermediate files) saved in a separate folder from other encounters. So for **each encounter** one should have:

- a separate folder for raw data (.wav files)
- a separate folder for raw cross-correlation
- a separate folder for results (Extracted TDOA tracks)

Note, most of these folders (apart from essential folders holding your data- i.e. .wav files) will be automatically created on a path you specify if they do not already exist.

2.3.2 Modify

First, specify paths, the array, and parameters for your application by modifying the following scripts:

- **Specify_Paths.m**: Specify folders where data are located and where results should be saved to. It is expected that you will be processing one encounter at the time (or if individual files / smaller range of files are processed, that these files are from the same encounter). See also 2.3.1 for information on the expected folder structure. The expected data format are .wav files. The expected name for .wav files is “xxx_yyyyMMdd_HHmmss_SSS.wav”, where “xxx_” can be any string (or none), and “yyyymmdd_HHmmss_SSS” part of the name specifies the date (year, month, day) and time (hours, minutes, seconds, and milliseconds (SSS)).

- **Array_info.csv:** This is where your hydrophone array sensor separation is specified. If adding a new array, enter the array information and sensor separation in a new row. Note, the name you give your array should then match the one you specify in the `Specify_Parameters4Xcorr.m`.

Important: The hydrophone sensor spacing should be specified in a continuous manner in meters from the first to the last hydrophone. For example: Assume your array is towed 300 m behind the boat, and consists of two sub-arrays that are separated by a 20 m separator, and each sub-array has three sensors that are separated by 1 m (Fig. 3.1). Further assume that your first sensor in each sub-array is placed right at the very beginning of a given sub-array, and assume the last sensor in each sub-array is placed right at the very end of a given sub-array. Then you enter the following values for hydrophones (each in a corresponding column): 0, 1, 2, 22, 23, 24. In practice, there is typically some separation between the beginning of the sub-array and the first sensor, and between the last sensor and the end of the sub-array (so measure and note sensor positions in a continuous manner). You will also have to enter the distance at which the array is towed behind the boat- this is a distance from the position of the GPS antenna, including the length of the tow cable, to the start of the first sub-array (or array if you only have one section with sensors). Carefully measure and note all sensor positions in a continuous manner. Insert -999 for fields where spacing is unknown (but the channel exists in the recordings), or if your number of sensors is less than 6.

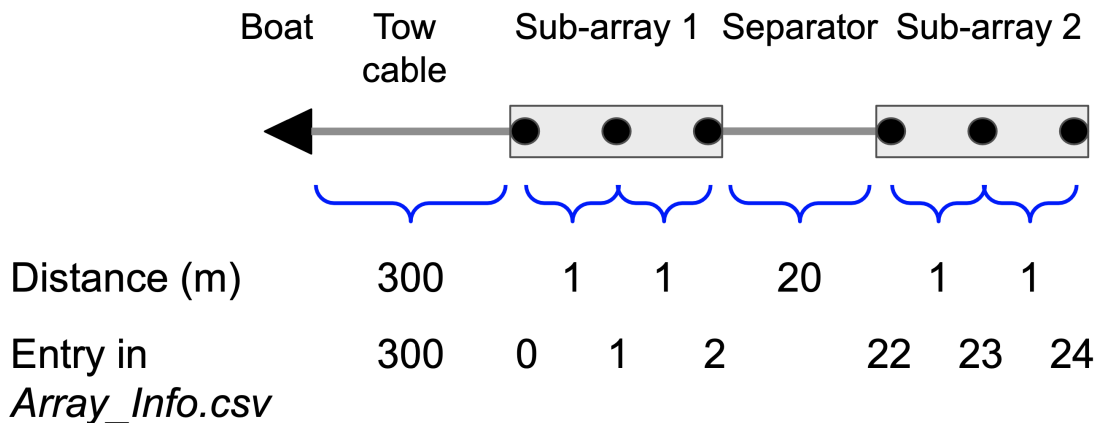


Figure 2.1: Diagram of the Array spacing and associated measurements. Each sub-array contains 3 sensors (black circles).

- **Specify_Parameters4Xcorr.m:** This is where you specify your array, encounter information, and settings for the cross-correlogram. Scroll down to change any parameters in the sections labeled “CHANGABLE:” as needed. The parameters are outlined in Chapter 5, Section 5.1, and documented in the function.
- **Specify_Parameters4Tracking.m:** This is where you specify parameters for measurement extraction and tracking. Scroll down to change any parameters in

the sections labeled “CHANGABLE:” as needed. The parameters are outlined in Chapter 5, Section 5.1, and documented in the function.

2.3.3 Run

Before running scripts below, your “Current Folder” must be navigated to the main folder of this package (then paths will get added automatically) OR add the main folder and subfolders to path manually. Either works!

Then run the package by running:

- 1) **A1_Compute_CrossCorrelograms.m**: This computes and saves the cross-correlogram based on your audio data. IMPORTANT: specify what signal type you want to be processing for (line 16) - choose either `clicks`, `whistles`, or `both`, depending on your application.
- 2) **A2_ExtractMeasurements_and_TrackTDOAs.m**: This extracts measurements and tracks TDOAs based on that and returns the extracted TDOA tracks. Results are also displayed as plots.

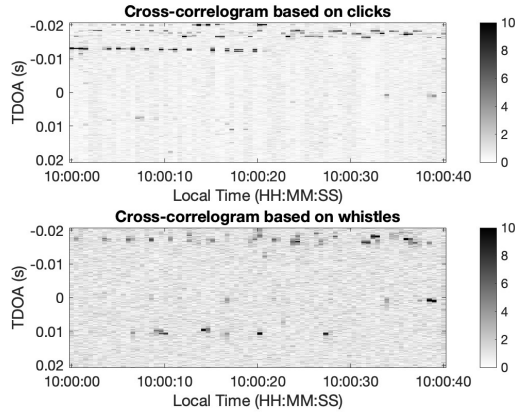
2.4 Output

The package outputs cross-correlograms, extracted TDOA tracks, plots, and saves the results. Specifically, the outputs that are saved in a `.mat` file are:

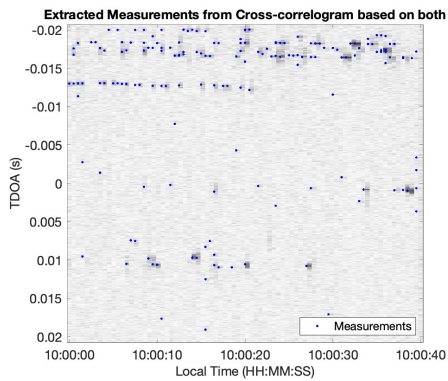
- 1) From **A1_Compute_CrossCorrelograms.m** the information saved in `<Encounter>_<whistles/clicks>_rawCrossCorrelogram_ALL.mat` is:
 - Cross-correlograms (`Rxy_envelope_ALL`) (based on envelopes of the generalized cross-correlation function) - matrix with a dimension of $N \times M$, where N is the number of time steps and M is the number of TDOAs.
 - Range of possible TDOAs (`lags`) - a vector of possible TDOAs for a given sensor separation.
 - Time vector (`t_serialdate`) - a time vector in datetime format (for more info type `help datetime` in Matlab’s command prompt)
 - Time vector (`t`) - a time vector in seconds from the beginning of the file/encounter.
 - Parameters used in the processing (`parameters`) - these are parameters specifying array and encounter information, and parameters used in cross-correlogram computation.
 - Parameters used specifically in processing of clicks and/or whistles (`param_signal`)
- 2) From **A2_ExtractMeasurements_and_TrackTDOAs.m** the information saved in `<Encounter>_Results.mat` is:
 - Extracted TDOA tracks (`Tracks`) - $1 \times M$ structure, where M is number of targets. For each target there are three fields: `time`, `time_local`, `tdoa`, where `time` refers to time in seconds from the start of the file/encounter, `time_local` refers to time in timedate format.

- Models used in GM-PHD-SA filter (`model`) - a structure array containing models and parameters for the filter.
- Parameters used for normalization, measurement extraction, and track extraction (`parameters_measure_tracking`).
- Parameters used in the cross-correlogram computation (`parameters`), (Comes from processing A1).
- Parameters used specifically in processing of clicks and/or whistles (`parameters_clicks` and `parameters_whistles` or `param_signal`), (Comes from processing A1).
- Scalars used for normalization of the cross-correlograms (`scalar_clicks` and `scalar_whistles` or `scalar`).

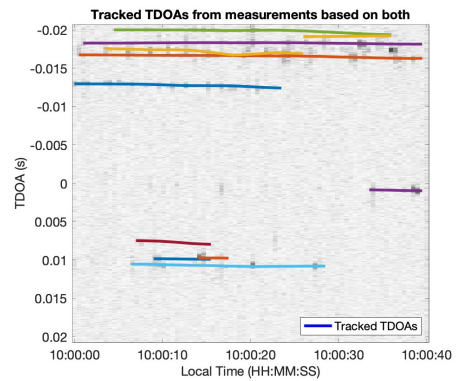
Example plots that are obtained as the output of this package are: plot of cross-correlogram(s) (2.2a); plot of extracted measurements (2.2b); and plot of extracted TDOA tracks (2.2c).



(a) Cross-correlograms.



(b) Extracted measurements.

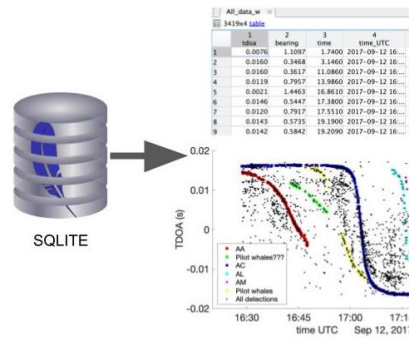


(c) Tracked TDOAs.

Figure 2.2: Example output plots for “TDOA tracking master” package.

Chapter 3

“Extract Pamguard detections” package



3.1 Package overview

This is a Matlab based package that contains code to extract Pamguard detections (clicks, whistles) from Pamguard SQLite3 database file to .csv and Matlab tables. It also computes Time-Difference-Of-Arrivals (TDOAs) for the detected signals (for a given sensor spacing).

This package was developed in Matlab, and tested on **Matlab version 2022a (9.12)**.

3.2 Package contents

The package contains the following functions, scripts, and files in the main folder:

- **A1_Extract_Tables_from_SQLite.m** - main script that extracts Tables from the Pamguard's SQLite3 database.
- **A2_Extract_Detections_Info.m** - main script that extracts Detected and Annotated (if available) vocalizations from Pamguard files.
- **Array_Info.csv** - a table specifying array spacing.
- **README.md** - readme document specifying package usage.
- **Specify_array_parameters.m** - specifies parameters for your encounter and array.
- **Specify_paths.m** - specifies paths to folders where data is located and results saved to.

The package contains the following folders in the main folder:

- 1) **./extract_detections_code/** - contains code to extract tables from SQLite3 database, and extract detections information from tables. Functions included are:

- `extract_detections_PAMcsv.m` - extracts relevant detection information from .csv tables into a Matlab table.
 - `extract_table_from_sqlite.m` - extracts a specified table from a SQLite3 database.
- 2) `./matlab-sqlite3-driver/` - Third party package that contains driver for SQLite3 database.
 - 3) `./Test_example/` - contains data to use for testing the package works correctly.

3.3 How to use

To use this package you must first install the driver for SQLite3 database. This is a third party package, for instructions see `README.md` in `./matlab-sqlite3-driver/README.md`.

As outlined in 1.4, the package expects a certain folder structure, that is for convenience outlined in 3.3.1.

After the driver installation, you must specify paths and array parameters (see Section 3.3.2). After this you can run the extraction and TDOA computation (see Section 3.3.3). The package outputs are outlined in Section 3.4.

3.3.1 Expected folder structure

It is expected that each encounter will be in a separate folder and have its results (and intermediate files) saved in a separate folder from other encounters. So for **each encounter** one should have:

- a separate folder for SQLite database
- a separate folder for extracted csv tables
- a separate folder for results (Pamguard detections)

Note, most of these folders (apart from essential folders holding your data- i.e. SQLite database) will be automatically created on a path you specify if they do not already exist.

3.3.2 Modify

First, specify the paths to folders and array parameters for your application by modifying the following scripts:

- **Array_info.csv**: This is where your hydrophone array sensor separation is specified. If adding a new array, enter the array information and sensor separation in a new row. Note, the name you give your array should then match the one you specify in the `Specify_array_parameters.m`.

Important: The hydrophone sensor spacing should be specified in a continuous manner in meters from the first to the last hydrophone. For example: Assume your array is towed 300 m behind the boat, and consists of two sub-arrays that are separated by a 20 m separator, and each sub-array has three sensors that are separated by 1 m (Fig. 3.1). Further assume that your first sensor in each sub-array is placed right at the very beginning of a given sub-array, and assume the last sensor in each sub-array is placed right at the very end of a given sub-array. Then you enter the following values for hydrophones (each in a corresponding column): 0, 1, 2, 22, 23, 24. In practice, there is typically some separation between the beginning of the sub-array and the first sensor, and between the last sensor and the end of the sub-array (so measure and note sensor positions in a continuous manner). You will also have to enter the distance at which the array is towed behind the boat- this is a distance from the position of the GPS antenna, including the length of the tow cable, to the start of the first sub-array (or array if you only have one section with sensors). Carefully measure and note all sensor positions in a continuous manner. Insert -999 for fields where spacing is unknown (but the channel exists in the recordings), or if your number of sensors is less than 6.

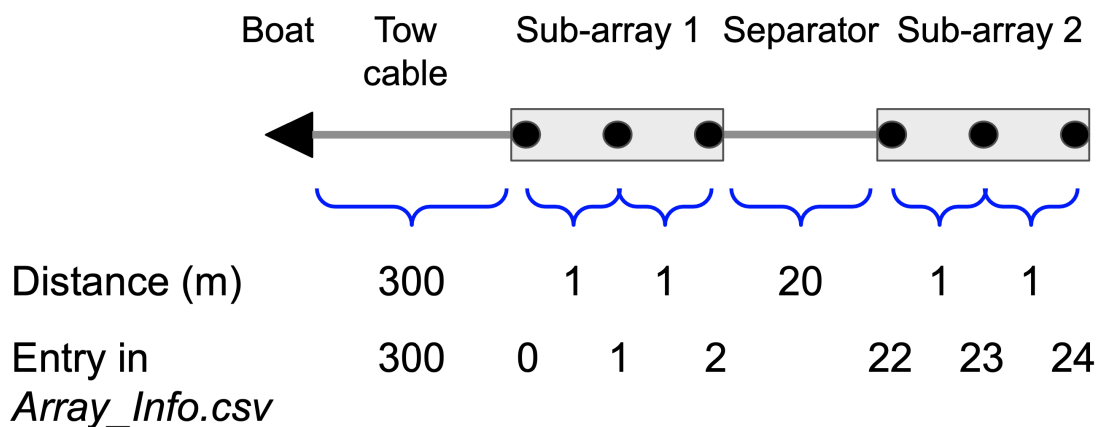


Figure 3.1: Diagram of the Array spacing and associated measurements. Each sub-array contains 3 sensors (black circles).

- **Specify_array_parameters.m** - This is where you specify your array and encounter information. Scroll down to change any parameters in the sections labeled “CHANGABLE:” as needed. The parameters are outlined in Chapter 5, Section 5.2, and documented in the function.
- **Specify_paths.m** - Specify folders where data is located and where results should be saved to. The expected folder structure is outlined in 3.3.1. The expected data format are `.sqlite3` database files that are produced by Pamguard (in here detections from whistle/click detectors are saved).

3.3.3 Run

Before running scripts below, your “Current Folder” must be navigated to the main folder of this package (then paths will get added automatically) OR add the main folder and subfolders to path manually. Either works!

Then run the package by running:

- 1) **A1_Extract_Tables_from_SQLite.m** - This extracts and saves specific tables (from click/whistle detectors) from `.sqlite3` database to `.csv` tables.
- 2) **A2_Extract_Detections_Info.m** - This extracts bearing information and computes TDOAs (for a given sensor separation) for all detections and annotated detections and saves them as a Matlab table. All detections and annotated detections are also displayed as plots.

3.4 Output

This package outputs `.csv` and Matlab tables (in `.mat` files- e.g., `<Encounter>_Extracted_Annotated<Clicks/Whistles>.mat`) of all and annotated detections (clicks/whistles) from Pamguard. All detections and annotated detections are also displayed as plots. Note, plots are not automatically saved.

Example plot that is obtained as the output of this package is shown in Fig. 3.2.

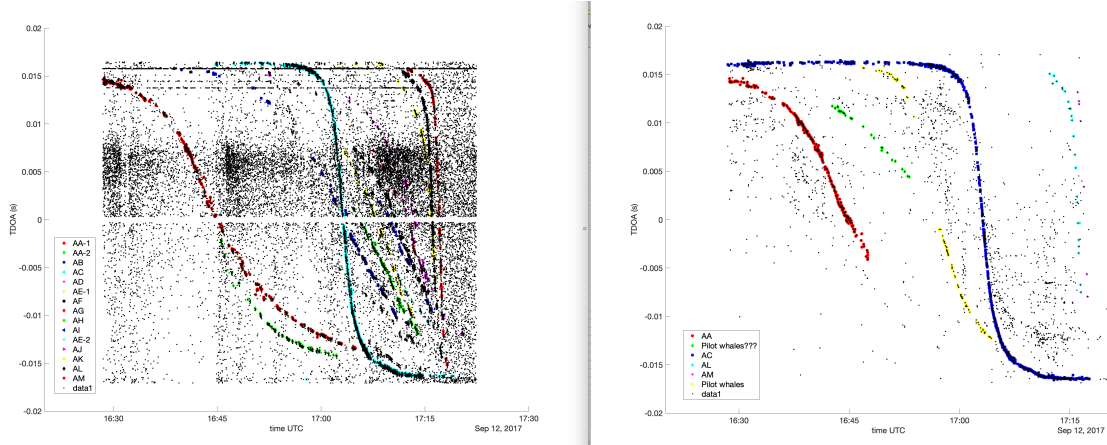
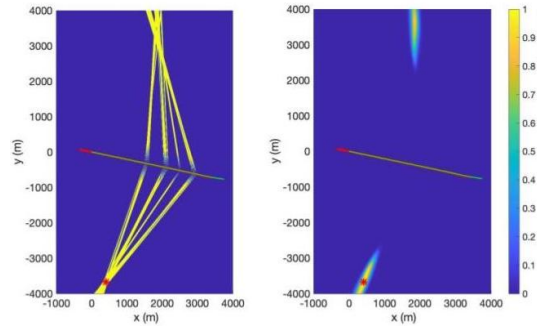


Figure 3.2: Example output plots for “Extract Pamguard detections” package. Raw Pamguard detections (black dots) as well as and user annotations (colored symbols) are shown.

Chapter 4



“Localization towed array” package

4.1 Package overview

This package contains code for the model-based localization of marine mammals from towed array data. Ambiguity surfaces (probabilistic indicators of source positions) [2, 3] are used to perform the localization.

Note, it is assumed that the boat moves in a straight line and no turns are present.

This package was developed in Matlab, and tested on **Matlab version 2022a (9.12)**.

IMPORTANT: Do not use earlier versions of Matlab since certain functions have changed behavior. For example, function `ismembertol` has different behavior between version 2022a and older versions, and the processing will not work correctly if using older versions. Also, earlier versions of Matlab do not have certain functions such as `clim` and `FontSize`, so these will not work. Also, **do not use Matlab version 2023b (23.2.0.2485118)**, since function `ismembertol` also changed behavior and the package will not work correctly.

4.2 Package contents

This package contains functions and scripts to perform model-based localization with ambiguity surfaces. You can either use simulated data or real data. The code for simulations is available in a folder `./Simulations/`. The rest of the package is for real data.

The package contains the following functions, scripts, and files in the main folder:

- **A0_GetGPSData.m** - main script that reads all relevant GPS data for a given encounter from a larger GPS database, and computes sensor positions for each time step.
- **A0_SetUp.m** - main script that reads all relevant data into the Matlab workspace,

selects TDOA tracks and relevant sensor positions that will be used to obtain final localization.

- **A1_Localize.m** - main script that localizes sources using a model-based ambiguity surface approach.
- **README.md** - readme document specifying package usage.
- **specify_parameters.m** - specifies parameters for towed array localization.
- **specify_paths.m** - specifies paths to folders where data is located and results are saved to.

The package contains the following folders in the main folder:

- 1) **./AS_Localization_code/** - contains code to compute ambiguity surfaces and provide localization based on that. Functions included are:
 - **computeAS.m** - computes an ambiguity surface (AS) for a given source based on the measured and modeled time difference of arrivals (TDOAs).
 - **findpeaks2D.m** - finds peaks in the ambiguity surface and computes their widths.
 - **get_xyrange.m** - limits the search range to a narrower area around the source location.
 - **GetGPSandPhonePositions.m** - creates a table with GPS data and sensor positions for a given time-frame of the encounter.
 - **localize_track.m** - localizes a selected TDOA track(s) and computes perpendicular distance between localized source and trackline.
 - **localizetracks.m** - localizes user selected TDOA tracks/track segments. It gives a visual plot of the localization and offers an option for keeping/discarding localization and localizing multiple tracks.
 - **M2LatLon.m** - converts position in meters to lat/lon.
 - **make_2Dgrids.m** - creates a 2-D grid and also outputs a structure of grid parameters.
 - **obtain_ambiguitysurface.m** - computes an ambiguity surface for a given source for a specified grid and surface parameters.
 - **plot_AS.m** - plots the final ambiguity surface, boat, array, and sensor positions.
 - **plot_tracks.m** - plots all and selected TDOA tracks against Pamguard detections (if available) or against cross-correlograms.
 - **select_tracks.m** - allows user to select which TDOA track / fragments they want to localize.
 - **simulate_array_pos.m** - simulates sensor positions assuming the boat moves in a straight line specified by the slope and intersect.
- 2) **./Simulations/** - code to perform localization with a model-based ambiguity surface approach using simulated data and scenarios.

- 3) `./Test_example/` - contains data to use for testing that the package works correctly. **IMPORTANT:** some of the test files were too big to be stored on GitHub. These are data for the folders "Crosscorrelograms" and "Raw_GPSandArray_info". Each of these folders contains README.txt that contains a link where the files can be downloaded. For convenience the link to the folder containing data is [this](#).

4.3 How to use

4.3.1 To use with simulated data

Go to folder `./Simulations/` and use `towedarray_LS_Simulations.m` for Simulation scenarios. Available simulations:

- Simulation 1: Stationary source, moving array, no noise on measurements
- Simulation 2: Stationary source, moving array, noisy measurements
- Simulation 3: Moving source, moving array, no noise on measurements - Incorporates Dilation of Surfaces
- Simulation 4: Moving source, moving array with more sensors, no noise on measurements
- Simulation 5: Two stationary sources, moving array, no noise

4.3.2 To use with real data

To use with Test data- refer to section 4.2, point 3).

For all other real data: This packages assumes that you have run the **"TDOA tracking master"** package first and that you have extracted the available TDOA tracks (see Chapter 2). Optionally, if you have Pamguard detections available, you should have run **"Extract Pamguard detections"** package (see Chapter 3) before running this current package.

As outlined in 1.4, the package expects a certain folder structure, that is for convenience outlined in 4.3.2.1.

First, specify paths and parameters you want to use for localization - see Section 4.3.2.2. After this, you can run the localization - see Section 4.3.2.3. The outputs are outlined in Section 4.4.

4.3.2.1 Expected folder structure

It is expected that each encounter will be in a separate folder and have its results (and intermediate files) saved in a separate folder from other encounters. So for **each encounter** one should have:

- a separate folder for Extracted TDOA tracks - i.e., where you stored results from "TDOA tracking master" package

- (Optional) a separate folder for Pamguard detections - i.e., where you stored results from “Extract Pamguard detections” package
- a separate folder for raw GPS data
- a separate folder for extracted table from GPS data
- a separate folder for Final localization results

Note, most of these folders (apart from essential folders holding your data- i.e. extracted TDOA tracks) will be automatically created on a path you specify if they do not already exist.

4.3.2.2 Modify

Before running the package, specify the paths and parameters for your application by modifying the following scripts:

- **specify_paths.m**: Specify folders where data is located and where results should be saved. The expected folder structure is outlined in 4.3.2.1. The expected data format are .mat files obtained from “TDOA tracking master” (e.g., <Encounter>_Results.mat and <Encounter>_<whistles/clicks>_rawCrossCorrelogram_ALL.mat) and optionally “Extract Pamguard detections” (e.g., <Encounter>_Extracted_Annotated<Clicks/Whistles>.mat) packages. If GPS and sensor depth information are available, these are expected to be in .csv format.
- **specify_parameters.m**: This is where you specify parameters for localization. Scroll down to change any parameters in the sections labeled “CHANGABLE:” as needed. The parameters are outlined in Chapter 5, Section 5.3, and documented in the function.

4.3.2.3 Run

Before running scripts below, your “Current Folder” must be navigated to the main folder of this package (then paths will get added automatically) OR add the main folder and subfolders to path manually. Either works!

Run the package by running:

- 1) [Optional] **A0_GetGPSData.m**: If real GPS data is available, this reads all relevant GPS data into the Matlab workspace and computes sensor positions. It saves results into a .csv table (at the location specified in **specify_paths.m**).
- 2) **A0_SetUp.m**: This reads all relevant data into the Matlab workspace. It also selects TDOA tracks and relevant hydrophone positions that will be used to obtain final localizations. **A0_SetUp.m** also generates figures that are used in the next step - **A1_Localize.m**. **Do not close these figures.**

- 3) **A1_Localize.m**: This localizes user selected group(s) of animals (TDOA track(s)). It requires user input on which track or which of the track fragments (since one track is often fragmented into separate track fragments) one wants to localize. This is interactive part where user is asked questions and is expected to interact through the commandline. The following questions are asked:
- Which track/tracks you want to localize? Look at Figure 1 and 2 for track numbers.
 - Do you want to keep this localization? (1 for yes, 0 for no)
 - Localize another group? (1 for yes, 0 for no)

Note, if there are not sufficient bearing changes in the selected tracks to achieve the localization (in question a.), then the user is prompted to try again. If the user wants to stop localizing, they need to choose 0 at question c. If the user wants to stop localizing at question a., then they should select something they localized already, then at the questions b. and c. select 0.

Results are saved into a table and contain localization estimates (Cartesian coordinates and latitude/longitude) and the perpendicular distance to the array for localized sources. Localized groups are also plotted. In case the process exists unexpectedly (e.g. computer crash, force quit) the temporary `TempResults.mat` is saved in the specified Results folder. Note, no other output is saved in this case and if the user wishes to keep this result, they must rename this file, otherwise it will be re-written next time `A1_Localize.m` is run.

4.4 Output

The package outputs:

- Output of Simulated scenarios are localization estimates and plots of the results.
- Output of Real data processing is a table containing localization estimates (in Cartesian coordinates and latitude/longitude formats) and the perpendicular distances to the array for each selected TDOA track (or fragments of the same TDOA track), and plots of the results. The variables that are saved in a `.mat` file (`Results_<Encounter>_StartTime_<First recording Date>_<First recording Time>_AnalyzeTime_<Today's Date>_<Time when saved>`) are:
 - Table containing localization information (`Loc_table`) - each source is in a separate row. The table contains the following columns:
 - * `TrackID` - selected track indices;
 - * `Loc_m` - localization estimate in Cartesian coordinates with respect to start of boat track (in m);
 - * `Loc_LatLong` - localization estimate as latitude and longitude (in decimal degrees);
 - * `Loc_m_dilated` - localization estimate from dilated AS in Cartesian coordinates with respect to start of boat track (in m);

- * `Loc_LatLong_dilated` - localization estimate from dilated AS as latitude and longitude (in decimal degrees);
- * `distance_m` - perpendicular distance from localization estimate (`Loc_m`) to the array (in m);
- * `distance_m_dilated` - perpendicular distance from localization estimate from dilated AS (`Loc_m_dilated`) to the array (in m).
- Final ambiguity surface (`AStotal`) - a 1 x M cell array where M is a number tracked sources/groups. Each cell is an A x B matrix where A is number of y coordinates, and B is number of x coordinates.
- Final ambiguity surface with dilation (`ASdilatetotal`) - a 1 x M cell array where M is a number tracked sources/groups. Each cell is an A x B matrix where A is number of y coordinates, and B is number of x coordinates.
- Final surface with intersecting hyperbolas (`AShyperbolas`) - a 1 x M cell array where M is a number tracked sources/groups. Each cell is an A x B matrix where A is number of y coordinates, and B is number of x coordinates.

Example plot that is obtained as the output of this package is shown in Fig. 4.1.

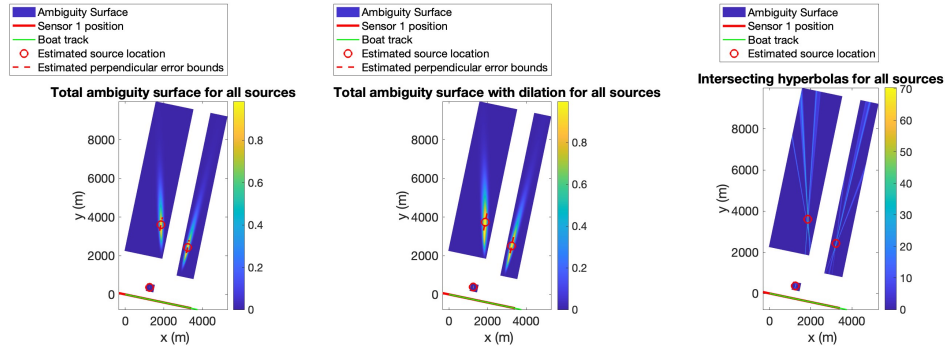


Figure 4.1: Example output plot for “Localization towed array” package.

Chapter 5

Description of changeable parameters

5.1 “TDOA tracking master” package

Parameters to change in this package can be found in `Specify_Parameters4Tracking.m` and `Specify_Parameters4Xcorr.m`. These are parameters in all of the sections labeled “CHANGABLE”.

Parameters in `Specify_Parameters4Tracking.m`:

- `parameters.lambda` - threshold above which the cross-correlation peaks will be extracted (default is $\sqrt{-2 * \log(0.001)}$).
- `parameters.excl_deg` - cut off for bearing in degrees- i.e. no bearing smaller than `parameters.excl_deg` degrees will be considered.
- `parameters.tmax` - maximum time (in s) to be considered in the noise sample-time from the beginning of the encounter until just before any significant sources occur (e.g. 1000s). NOTE: the longer the sample the better the background estimation.
- `parameters.min_dist` - minimum distance (in seconds) between two neighbouring cross-correlation peaks. NOTE: minimum distance (in seconds) should be LESS than $2 * \text{max possible TDOA}$ (determined by your sensor separation- $2 * \text{max}_{\text{tdoa}} = 2 * d/c$ - specified in `Specify_Parameters4Xcorr.m`). Default for false killer whale whistles is 0.0031 s.
- `parameters.min_tl` - minimum track length criteria- a TDOA track must be at least `parameters.min_tl` long in order to become a true track. For example: 3 s.
- `movingmeanlength_s` - the length of the moving average (for track smoothing) - if no track smoothing & interpolation are desired then set `movingmeanlength_s=0`;
- `parameters.saveplotsofresults` - save result Plots - specify if you want to save plots of your results (1=yes, 0 = no).

Parameters in `Specify_Parameters4Xcorr.m`:

- `parameters.year` - survey year.
- `parameters.arrayname` - array name.
- `parameters.encounter` - encounter name.
- `parameters.c` - speed of sound - should be a scalar, it will be used for limiting the possible TDOAs (min & max), so it can be approximate.
- `parameters.channels` - which two channels of your recordings you want to cross-correlate. Select two that are furthest apart.
- `parameters.method` - the method for generalized cross-correlation (GCC): 'scc'/'phat'/'scot' (see `gcc.m` for more info)
- `parameters.window_length_s` - window length (in s) over which the cross-correlation is computed (default for false killer whales is 1 s, but this is problem dependent).
- `parameters.overlap` - proportion of overlap between consecutive windows (default is 0.5 - i.e. 50%).
- `varargout{.}.freq_filter` - bandpass filter cut-off frequencies [lower, upper] (in Hz). !! **NOTE** !! - only change the `varargout{.}.freq_filter` in the `switch` statement and make sure you change it in the appropriate section ('clicks'/'whistles'/'both'). Some defaults for **clicks**: [8000,30000] - for false killer whales; [5500,20000] - for sperm whales; [10000,80000]- for Rissos dolphins; [10000,50000] - for rough toothed dolphins. Some defaults for **whistles**: [2500,12000] - for false killer whales; [3300,28000]- for rough toothed dolphins.
- `parameters.plotcrosscorr` - plot cross-correlogram (0 = no, 1 = yes).
- `parameters.saveworksp` - save cross-correlogram & parameters (0 = no, 1 = yes).

5.2 “Extract Pamuard detections” package

Parameters to change in this package can be found in `Specify_array_parameters.m`. These are parameters in all of the sections labeled “CHANGABLE”.

Parameters in `Specify_array_parameters.m`:

- `parameters.year` - survey year.
- `parameters.arrayname` - array name.
- `parameters.encounter` - encounter name.
- `parameters.c` - speed of sound - should be a scalar, it will be used for limiting the possible TDOAs (min & max), so it can be approximate.

- `parameters.channels` - which two channels of your recordings you want to cross-correlate. Should match whatever you have chosen in “TDOA tracking master package” if using that.

5.3 “Localization towed array” package

Parameters to change in this package can be found in `Specify_parameters.m`. These are parameters in all of the sections labeled “CHANGABLE”.

Parameters in `Specify_parameters.m`:

- `boatspeed_kts` - boat speed in knots. **!! NOTE !!** - only change if simulating GPS positions.
- `BA_params.mb` - line gradient of the line along which the boat moves. **!! NOTE !!** - only change if simulating GPS positions.
- `BA_params.b` - intersect of the line along which the boat moves with y axis. **!! NOTE !!** - only change if simulating GPS positions.
- `AS_params.sig` - sigma (standard deviation) for the Gaussian (default is 0.003).
- `AS_params.sig_hyperbolas` - standard deviation for plotting intersecting hyperbolas (for visual assessment of how they are crossing), (default is 0.00003).
- `minxrange` - minimum x range to include in the ambiguity surface computation (default is `min(boat_pos(:,1)) - 400`: i.e. - 400m to include the array behind the boat).
- `maxxrange` - maximum x range to include in the ambiguity surface computation (default is `max(boat_pos(:,1)) + 2000`).
- `AS_params.yrange` - y range in m [min, max]- since boat track will be rotated around x-axis, and localization is bi-ambiguous specify positive y-range or in case sources are near trackline can add a few m on the other side- e.g. -300m.
- `AS_params.dx=10` - grid step size in m (resolution) in x direction (default is 10 m).
- `AS_params.dy` - grid step size in m (resolution) in y direction (default is 10 m).
- `AS_params.mean_horiz_swimspeed` - mean horizontal swim speed [m/s] (default for false killer whales is 0.5).
- `AS_params.bearing_cutoff` - cutoff criteria for tracks around the beam - this will affect which tracks you consider for localization (default is 60).

Chapter 6

Troubleshooting

6.1 Error using Specify_Parameters4Tracking

If you see the following message in the command prompt - Fig. 6.1, then the minimum distance you have selected for the neighboring peaks in the cross-correlation function (`parameters.min_dist`) is bigger than what is physically possible for your array configuration. This means it exceeds twice the maximum possible TDOA, which is determined based on sensor separation `d` in `Specify_Parameters4Xcorr.m`. As the message says, you need to either decrease the `parameters.min_dist` or you need to process sensors that have larger separation. I recommend to always use the largest separation that is available for your array.

```
>> A1_Compute_CrossCorrelograms
Computing cross-correlograms...
Computing clicks cross-correlogram for file test_20220826_100000_000.wav
Computing whistles cross-correlogram for file test_20220826_100000_000.wav
>> A2_ExtractMeasurements_and_TrackTDOAs
Error using Specify_Parameters4Tracking
The chosen parameters.min_dist- i.e. minimum distance between
two neighboring peaks in cross-correlation is BIGGER than what
is physically possible - i.e. it exceeds 2 x maximum possible
TDOA determined based on sensor separation d in
Specify_Parameters4Xcorr.m. EITHER decrease the min_dist (in
Specify_Parameters4Tracking.m) OR choose to process sensors with
bigger separation (in Specify_Parameters4Xcorr.m).

Error in A2_ExtractMeasurements_and_TrackTDOAs (line 64)
[parameters_measure_tracking]= Specify_Parameters4Tracking(parameters.signal_type,parameters);

>>
```

Figure 6.1: Error message when `min_dist` is too large for the specified array.

6.2 I cannot find results- .mat files

In case you have trouble finding `.mat` files in a given folder, check how your computer interprets them. In some cases these files might appear as Microsoft access or another file format.

Bibliography

- [1] Pina Gruden, Eva-Marie Nosal, and Erin Oleson. Tracking time differences of arrivals of multiple sound sources in the presence of clutter and missed detections. *The Journal of the Acoustical Society of America*, 150(5):3399–3416, 2021.
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- [3] Yvonne M Barkley, Eva-Marie Nosal, and Erin M Oleson. Model-based localization of deep-diving cetaceans using towed line array acoustic data. *The Journal of the Acoustical Society of America*, 150(2):1120–1132, 2021.